

Question Paper

Exam Date & Time: 21-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
Nanotechnology and Applications [PHY_4345]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Assess the formation of energy bands in solids using the concept of overlapping atomic orbitals. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Explain the electrical properties of nano materials (3)
Course Outcome: "CO1" / Cognitive Level: "Understand"
C) Relate why the optical properties of quantum dots are size-dependent? (2)
Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Identify the principle of lithography? mention the types of nanolithography. Explain photo nanolithography with neat diagram. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Explain with suitable examples how Solvothermal synthesis can be tuned to control nanomaterial morphology. (3)
Course Outcome: "CO2" / Cognitive Level: "Understand"
C) Identify which method is suitable for preparing quantum dots and state the reason for your choice. (2)
Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Find how Bragg's Law is applied in determining the crystal structure of an unknown sample using X-ray diffraction (XRD) with neat diagram (5)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Apply the working principle of EDAX/EDS to explain how purity of the samples can be found for prepared nano materials. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Identify how Atomic Force Microscopy (AFM) can be used to measure surface roughness at the nanoscale. (2)
Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Apply your understanding of nanocomposite structures to assess how adding graphene sheets to a polymer matrix improves its mechanical and electrical performance. Support your answer with examples of practical applications (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Use the concept of fullerene structure to predict how changing the size (number of carbon atoms) affects its electronic and chemical properties. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Illustrate working principle of hybrid solar cells to design an energy-harvesting system for wearable electronic devices. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Use your knowledge of nano-sensors to propose how they could be applied in medical diagnostics for early disease detection. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_4345:Nanotechnology and Applications

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_4345:Nanotechnology and Applications

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	8(16%)
3	Application	10	42(84%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)